Statistics

Branch of Maths which deals with data

collection → analysis → Interpret → Presentation

Data → Statistics → Decision

Types of data

Quantitative

- Discrete
  - Exactly data in 1, 2, 3, 4 ...
- Continuous
  - Symbols: Height, Weight
  - 1', 1", 3m
  - Range

Qualitative

- Nominal
  - Values which don't have any order in them
  - Example: we know Red, Green, Yellow, Blue
  - Friends > English ??
  - Hindi > English
- Ordinal
  - Values which have order in them
  - Size: S, M, L, XL, XXL, XXXL
  - XS < S < M < L < XL < XXL

Questions: Classify the below in type of data

1) Name of students in class : Nominal
2) Size of shirts : Ordinal
3) Height of students : Continuous
4) Ordinal of City : Nominal (Hindi / Tel)

Types of statistics

1) Descriptive statistics
   - Describing the data through statistics

eg Reporting a trend of sales of a company

2) Inferential statistics
   - Use data and inferences from the data

eg

→ Inference: Use as a sharp growth in sales in 2023

Inference: People from 20-30 yrs of age, which shows for larger have confirmed to other age groups

Dive-deep into descriptive statistics

4 Measures :

1) Measure of frequency : The number of times a particular data

Can you represent India's population height in a chart

2) Measure of central tendency :

Who are taller, Indians or Americans??

Americans are taller

Are all americans are taller than us??

Avg. Americans height is more than Avg. Indian height

Indian height	American height
5'4" → 5'3 Male	5'9 Male
5'1 Female	5'3" Female

Central tendency

by below 1 value represent a population, its called central Tendency

Three central Tendencies :

1) Mean → Average
2) Median → Middle value in a population
3) Mode → Most frequent value

Q. Which one is the best central Tendency!

Ans → It depends

Class A :

Marks obtained	/100
0 → 5	
5 → 10	
10 → 15	
15 → 20	
20 → 25	
25 → 30	
30 → 35	

Class B

Marks obtained	/100
0 → 10	16
10 → 20	18
20 → 30	16
30 → 40	16
40 → 50	16
50 → 60	16
60 → 70	18

Avg marks : $\frac{\text{sum of all}}{\text{no. of students}}$

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A > B

Avg is not a good central Tendency in this case

Median → $\boxed{5 < 13.5}$ → is a good central Tendency

Avg → If there is an outlier in the data we prefer median over mean

In case of no outlier, we prefer mean over median

Mode → is used for categorical data

Most of the population is 'male'

→ Mode of the data

Gender: Male, Female, Male, Male, Male, Male

→ 85% are male, 15% are female

Most frequently occurring data pt. is called mode

Mathematical formulas of mean/median/mode.

Mean = $\frac{\text{Sum of all values}}{\text{No. of values}}$

in statistics

Mean = $\boxed{\frac{\sum f \cdot x}{\sum f}}$

'Σ' → sum

f → frequency of values

x → value

10, 20, 10, 15, 15, 10

Mean = $\frac{80}{6}$ = 13.33

x	f	f·x
10 →	3	30
15 →	2	30
20 →	1	20
$\sum f x$		80

$\sum f = 6$

$= \frac{80}{6} = 13.33$

Q - Calculate the mean of the following marks :

12, 15, 15, 29, 30, 21, 30, 30, 15, 19, 19, 15, 20, 20, 16, 21, 23, 29, 23, 21

20.8 is the right answer

Median : [For odd number of data pts]:

$\boxed{\left(\frac{n+1}{2}\right)^{th} \text{ term}}$

1, 2, 1, 4, 3, 5, 8, 6, 9

1st step : Arrange them in ascending order

1, 1, 2, 3, 4, 5, 6, 8, 9 — 9 data pts

$\left(\frac{n+1}{2}\right)^{th} \rightarrow \left(\frac{9+1}{2}\right)^{th}$ term = 5th term

Median = 4

For even number of data pts

$$\boxed{\frac{n^{th} \text{ term} + \left(\frac{n}{2}+1\right)^{th} \text{ term}}{2}}$$

1, 1, 2, 3, 4, 5, 8 = 8

$\frac{n^{th} \text{ term} + 5^{th} \text{ term}}{2} = \frac{3+4}{2} = \boxed{3.5}$

Q - Age of 9 boys in years are :

5, 2, 8, 6, 3, 15, 13, 12, 10

what is the median age.

1st step : Arrange in ascending order

2, 3, 5, 6, 8, 10, 12, 13, 15

Median = $\left(\frac{n+1}{2}\right)^{th}$ term

= 5th term = 8

Q - 10, 2, 5, 8, 12, 16

find the median!!

1st step : 2, 5, 8, 10, 12, 16 — 6 values

$$\frac{\frac{n}{2}^{th} \text{ term} + \left(\frac{n}{2}+1\right)^{th} \text{ term}}{2}$$

$= \frac{8+10}{2} = 9$

3.) Measures of position :

Percentile

Rahul is the tallest person in India

So he has the top position

Student from is India

So he has the last position